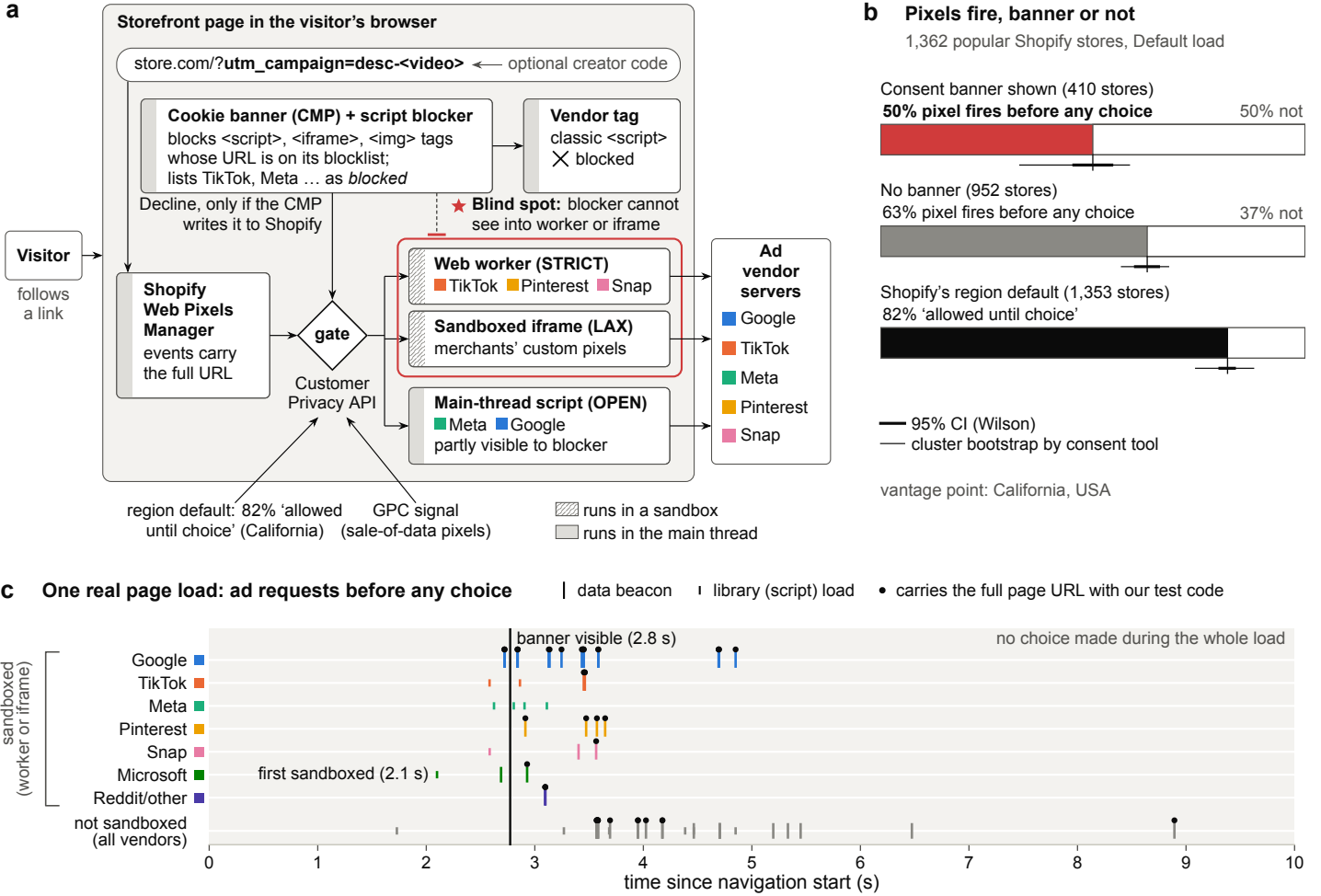

a
Storefront page in the visitor's browser
store.com/?utm_campaign=desc-<video>
optional creator code
Cookie banner (CMP) + script blocker
blocks <script>, <iframe>, <img> tags whose URL is on its blocklist; lists TikTok, Meta … as blocked
Vendor tag
classic <script>
blocked
Decline, only if the CMP writes it to Shopify
Blind spot: blocker cannot see into worker or iframe
Visitor
follows a link
Shopify Web Pixels Manager
events carry the full URL
gate
Customer Privacy API
Web worker (STRICT)
TikTok Pinterest Snap
Sandboxed iframe (LAX)
merchants' custom pixels
Main-thread script (OPEN)
Meta Google
partly visible to blocker
Ad vendor servers
Google
TikTok
Meta
Pinterest
Snap
region default: 82% 'allowed until choice' (California)
GPC signal (sale-of-data pixels)
runs in a sandbox
runs in the main thread
b
Pixels fire, banner or not
1,362 popular Shopify stores, Default load
Consent banner shown (410 stores)
50% pixel fires before any choice
50% not
No banner (952 stores)
63% pixel fires before any choice
37% not
Shopify's region default (1,353 stores)
82% 'allowed until choice'
95% CI (Wilson)
cluster bootstrap by consent tool
vantage point: California, USA
c
One real page load: ad requests before any choice
data beacon
library (script) load
carries the full page URL with our test code
banner visible (2.8 s)
no choice made during the whole load
sandboxed (worker or iframe)
Google
TikTok
Meta
Pinterest
Snap
Microsoft
Reddit/other
first sandboxed (2.1 s)
not sandboxed (all vendors)
0
1
2
3
4
5
6
7
8
9
10
time since navigation start (s)